MEDBHASHA · ROUND 2

Clinical notes, understood in Bangla.

Technical report: entity extraction, clustering and retrieval-grounded question answering over Bangla clinical notes.

Team · Train Test Panic | Seed 42 | No external APIs | No web retrieval

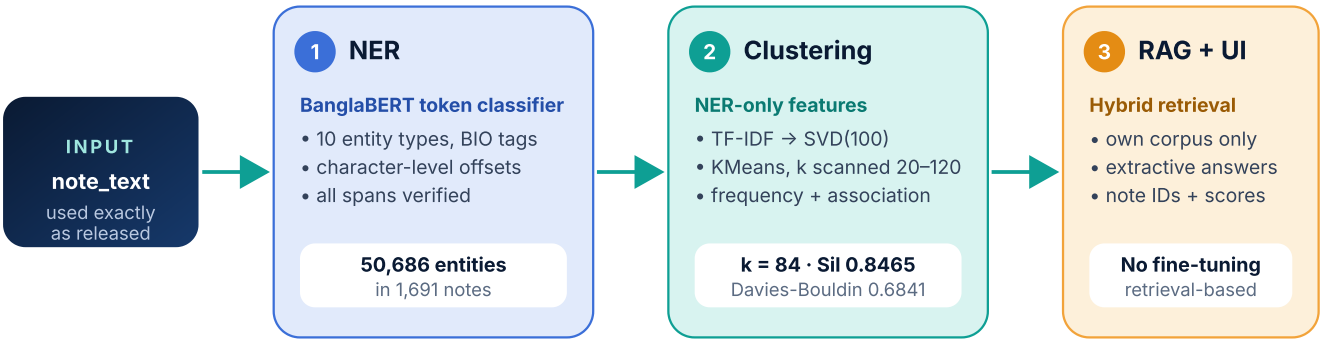
0.908	50,686	0.8465	k = 84
NER F1 · DEV	ENTITIES EXTRACTED	SILHOUETTE (COSINE)	CLUSTERS
vs 0.844 CRF baseline	across 1,691 notes	Davies-Bouldin 0.6841	scanned k = 20–120

01 · OVERVIEW

One notebook, three stages, fully reproducible

The whole solution lives in a single notebook, `notebooks/final_notebook.ipynb`. It runs three stages in sequence: **NER** turns raw note text into typed entities, those entities feed **clustering and relationship analysis**, and the same annotated corpus powers a **RAG** question-answering interface.

PIPELINE · SINGLE NOTEBOOK · SEED = 42



Every downstream stage consumes the NER output or the annotated corpus — nothing is fetched from outside.

Design principles

Self-contained

No external APIs and no web retrieval. The RAG index holds the team's own annotated corpus only.

Faithful to input

Test notes are used exactly as released — no rewriting, so every offset maps to the original text.

Deterministic

Seed fixed at 42, deterministic KMeans/SVD and pinned `requirements.txt`.

Results at a glance

STAGE	METHOD	HEADLINE RESULT
1 · NER	BanglaBERT token classification, BIO tags, 10 entity types	**F1 0.908** on dev (CRF 0.844); 50,686 entities in 1,691 notes
2 · Clustering	Entity-only TF-IDF, SVD(100), KMeans	**k = 84**, silhouette 0.8465, Davies-Bouldin 0.6841
3 · RAG + UI	Hybrid retrieval over own annotated corpus	Extractive answers with note IDs, passages, scores, concept coverage

02 · NAMED ENTITY RECOGNITION

BanglaBERT beats the CRF baseline on every metric

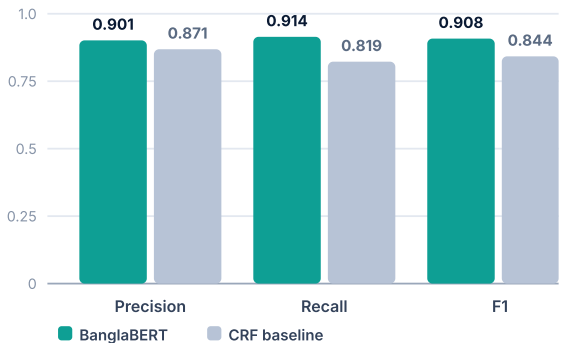
0.908
F1 · BANGLABERT
entity-level, exact match

+0.064
F1 VS CRF
0.844 → 0.908

300
ANNOTATED NOTES
14,452 tokens

10
ENTITY TYPES
BIO tagging

Dev results (42 notes)



Gain over the CRF baseline: precision +0.030, recall +0.095, F1 +0.064. The biggest win is recall.

Model & training

- **csebuetnlp/banglabert**, fine-tuned for token classification
- BIO tags over 10 entity types; first-sub-word labelling
- lr = 5e-05, batch = 16, AdamW with linear warm-up
- Final model retrained on all 300 notes for **23 epochs**

Data & split

- 300 self-annotated notes, 14,452 tokens
- Dev = 42 notes, stratified by Round-1 id prefix

Exact character offsets

The tokeniser keeps `(start, end)` for every token. An entity spans the **first token's start** to the **last token's end** on the unmodified `note_text`.



Code points, 0-based, end exclusive. Every span passes `note_text[start:end] == entity_text`.

Output

50,686 ENTITIES

1,691 NOTES

About 30 entities per note on average — dense enough to drive clustering, association mining and retrieval downstream.

Known limitation — domain shift. Round-1 training data is narrative prose, while many test notes are telegraphic `C/O … Dx: … Rx:` style. Some performance drop on those notes is expected.

03 · CLUSTERING & 04 · RELATIONSHIP ANALYSIS

Tight, well-separated clusters from entities alone

Features come **only from NER output** — a typed entity bag plus word unigrams — so clustering stays clinically meaningful and independent of raw text noise.

Typed entity bag
+ word unigrams

→

TF-IDF weighting
Dx ×3 · Med ×2 · Sym ×1.5

→

TruncatedSVD
100 components

→

KMeans · k = 84
within 5 % of best silhouette

Cluster quality

Silhouette (cosine) higher is better — 0.8465

Davies-Bouldin lower is better — 0.6841

Computed on our own data. Hidden-label NMI / ARI are computed by the organisers.

Choosing k

- Scanned **k = 20 to 120**
- Picked the **smallest k within 5 %** of the best silhouette — fewer, more interpretable clusters at almost no quality cost
- Result: **k = 84**
- Full scan saved in `predictions/analysis/cluster_k_selection.csv`

Frequency & relationship analysis

All outputs are in `predictions/analysis/` :

Frequency
Tables and charts per entity type.

Co-occurrence
Entity-type co-occurrence and an entity network.

Associations
Diagnosis ↔ medication / symptom / test with support, lift and PMI; diagnosis × medication heat-map; cluster profiles.

TOP PREDICTED DIAGNOSES

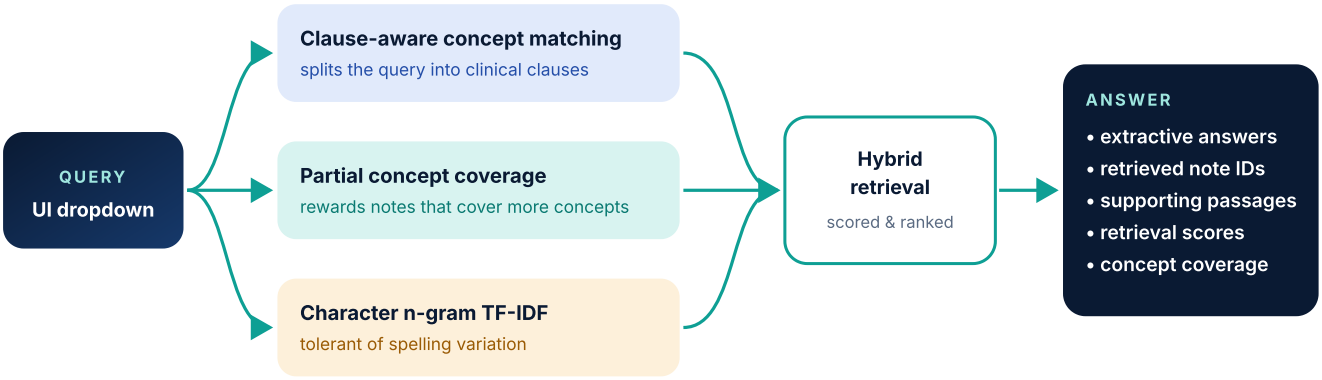
- ডায়াবেটিস — Diabetes
- উচ্চ রক্তচাপ — Hypertension
- টনসিলাইটিস — Tonsillitis
- জল বসন্ত — Chickenpox
- স্ক্যাবিস — Scabies
- অ্যাটোপিক ডার্মাটাইটিস — Atopic dermatitis
- epigastrium — as predicted
- অ্যালার্জিক রাইনাইটিস — Allergic rhinitis

TOP MEDICATIONS

- প্যারাসিটামল — Paracetamol
- সিটিরিজিন — Cetirizine
- পেনিসিলিন — Penicillin
- ওআরএস — ORS
- অ্যামোক্সিসিলিন — Amoxicillin
- ক্যালামাইন লোশন — Calamine lotion
- আই ড্রপ — Eye drops
- অ্যাসপিরিন — Aspirin

Grounded answers, with receipts for every claim

Retrieval-based, with no fine-tuning. The index contains the team's own annotated corpus only, so every answer can be traced back to real notes.



Why this design works

- **No hallucination path:** answers are extracted or aggregated from retrieved notes, never free-generated.
- **Test-result queries** use the annotated `TEST_RESULT` / `TEST` entities directly.
- **Auditable:** each answer lists note IDs, passages, scores and concept coverage.

One retrieval function

The dropdown UI calls the **same retrieval function** that produced the saved fixed-query outputs, so what judges see in the UI matches the submitted results exactly.

Reproducibility checklist

- ✓ **requirements.txt** provided
- ✓ Fixed seed = **42** everywhere
- ✓ Deterministic KMeans and SVD
- ✓ Single notebook end to end
- ✓ No network use, except downloading the public BanglaBERT checkpoint and normaliser
- ✓ Test notes used exactly as released

Summary. A fine-tuned BanglaBERT tagger (F1 0.908) feeds both an entity-only clustering (k = 84, silhouette 0.8465) and an auditable hybrid-retrieval QA system — all deterministic, offline and built solely on our own annotations.